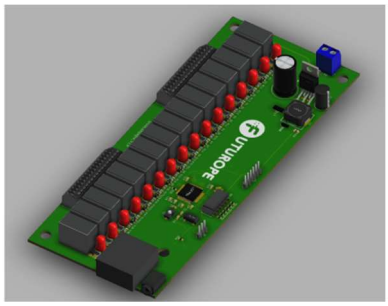
Board 16 Relays :



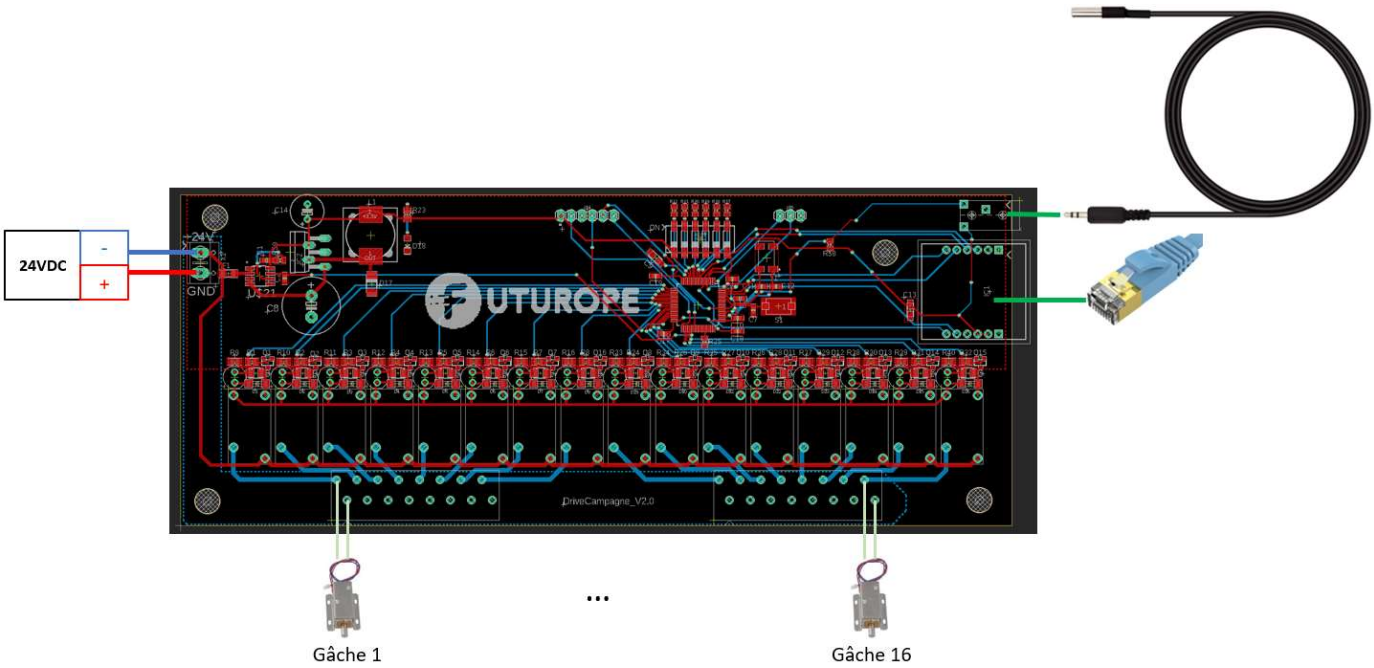
Features :

- **STM32F411RE** Microprocessor, 32-Bit ARM Cortex
- **W5500** Ethernet module
- **16 Relays** Output
- **1x Jack** connector for temperature sensor
- **ACS711** current sensors
- **2x16 Terminals** fast connector 24 AWG

Specification :

- DC Current **100µA**/MHz
- Clock speed up to **100** MHz
- **512KB** Flash Memory
- **128KB** SRAM
- Ethernet **802.3 10/100 Mbit/s**
- **Relays outputs**: 16 x 3A maximum (Normally Open)
- **Input Voltage**: (Vin) 16-30VDC
- **Dimension**: 200mm x 80mm x 35mm
- **Weight**: 250 g

Wiring :



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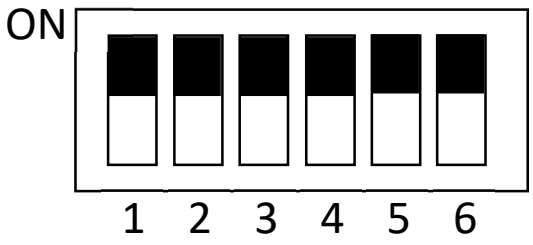
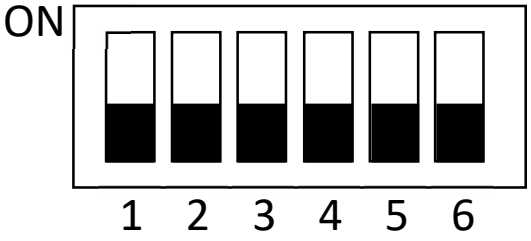
Hardware Addressing :

IP address = 172.16.10. + DIP SWITCH

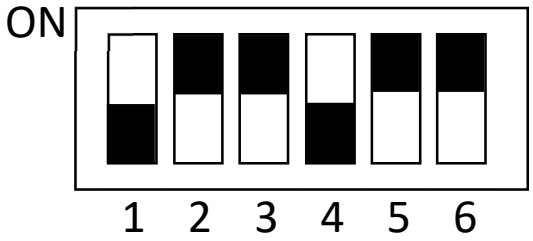
The IP address ended by 0 or 1 are forbidden. Some examples for better understanding :



IP Adress = 172.16.10.63

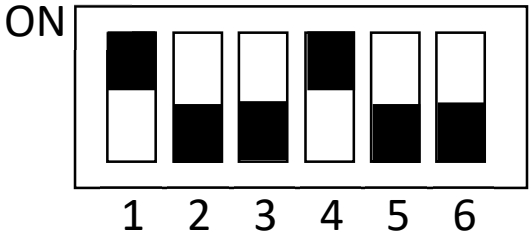


IP Adress = 172.16.10.2



lastDigit = $2^{\wedge 1} + 2^{\wedge 2} + 2^{\wedge 4} + 2^{\wedge 5}$

IP Adress = 172.16.10.54



lastDigit = $2^{\wedge 0} + 2^{\wedge 3}$

IPAdress = 172.16.10.9

Notice that 1 and 2 values for DIPSWITCH are forced in the firmware to be 172.16.10.2 IP address. Be careful to have no duplicate in the IP address (both won't work).

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Communication :

The computer talk to the device thanks to UDP protocol, the specifications are as follow :

UDP Socket = 12345

-UDP Message-	-Details-
FF0101	Relay 1 ON command (ASCII encoding)
FF0100	Relay 1 OFF command (ASCII encoding)
	...
FF1600	Relay 16 OFF command (ASCII encoding)
FF1601	Relay 16 ON command (ASCII encoding)
FFE000	All relays OFF command (ASCII encoding)
FFE001	All relays ON command (ASCII encoding)
FF0000	Get relays status (return 2 Bytes)
FFF000	Get temperature value (only available with sensors relay board option)
FFC000	Get current value (only available with sensors relay board option)

After a command an acknowledgment is sent from the device to the computer, composed as follow :

ACK = BOOL;COMMAND

BOOL = 1 command is a success

BOOL = 0 command is a failure

A failure could be due to a hardware issue. For example the relay is active but 0Amp is consumed by the electric charge.

For all details, source code is available on github : https://github.com/eporutuf/16_relay_board/blob/master/src/main.cpp

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